
AEGIS: Adaptive Emergency Grid for Incident Steering

An Intelligent Emergency Response & Resource Optimization Platform

Emergency dispatch is not primarily a routing problem; routing is a solved commodity. The hard problems are allocation under scarcity, stability under continuous change, and accountability for decisions affecting survival. AEGIS is a sharded, event-driven backend built around those three.

Incidents are accepted durably on a region-partitioned event log, then triaged by a versioned additive model that emits a per-term explanation rather than an opaque rank. Allocation runs in two tiers over one shared cost function: a sub-millisecond greedy admission decision, so no caller waits for a solver, and a budgeted epoch that solves each regional shard *exactly*. The key step, requirement expansion, converts an NP-hard generalized assignment problem into a rectangular linear assignment problem solvable optimally by Jonker–Volgenant; scarcity, hospital capacity, coverage reserve and plan stability enter as cost terms, refined by anytime Large Neighbourhood Search. Re-optimisation is debounced and change-triggered, with switching costs inside the objective so crews are never rerouted for marginal gain.

Conflict-freedom rests on three independent layers: single-writer sharding, TTL reservation leases with all-or-nothing group acquisition, and optimistic concurrency backed by a database constraint. Dependencies degrade through a defined ladder rather than failing.

Evaluated on a replayed 12-hour national surge against a nearest-available baseline: mean response -14% , p90 -39% , p99 -25% , severity-weighted -18% , incidents never reached -18% , utilisation $+37\%$, and zero resource conflicts.